

Math 340: Elementary Matrix and Linear Algebra

FINAL EXAM

Tuesday, December 17th, 2024, 7:45am-9:45am

Circle your Instructor and your TA:

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Name: _____ Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 120-minute exam. It consists of 10 problems for a total of 100 points; the exam is 8 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- If making multiple attempts at a solution, indicate clearly which attempt you'd like to be graded by crossing out the other answers. Multiple attempts will not be graded.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (in x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication and dot product unless otherwise noted.
- Additional notation reminders: O denotes the zero matrix, and I_n denotes the $n \times n$ identity matrix. $\text{tr}(A)$ denotes the trace of a matrix A .

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (7 points) Consider $T : P_2 \rightarrow P_1$ by $T(p(x)) = p(0) + 2p'(x)$. You can assume that T is a linear transformation.

Find $M_{DB}(T)$, where $B = (5, x^2 + 3, x)$ and $D = (1, x)$. Show all work.

(Hint: your final matrix will be 2×3).

2. (7 points total) Consider the subspace W of $\mathbb{R}^4$ spanned by the set

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ 6 \end{bmatrix} \right\}$$

a. (3 points) What is the dimension of W ? Show your work, and briefly justify your answer.

b. (2 points) For which d is W isomorphic to P_d ? Justify your answer in one sentence.

c. (2 points) Is W isomorphic to M_{22} ? Justify your answer in one sentence.

3. (20 points, 2 points each) Fill in the bubble next to the correct answer. Justification is not necessary.

a.) Let A be an $n \times n$ matrix with $\text{rank}(A) = n$. If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a set of linearly independent vectors in $\mathbb{R}^n$, then $\{A\mathbf{v}_1, \dots, A\mathbf{v}_k\}$ is a set of linearly independent vectors in $\mathbb{R}^n$.

- ☐ True
☐ False

b.) Let A be an $n \times n$ matrix. If A^2 is invertible, then A is invertible.

- ☐ True
☐ False

c.) Let V be a finite-dimensional vector space where $\dim(V) = n$. Let B and D be ordered bases for V , and let $T : V \rightarrow V$ be a linear operator on V . Then $\det(M_B(T)) = \det(M_D(T))$.

- ☐ True
☐ False

d.) Consider P_1 with the following inner product: $\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$. Using this inner product, x is a unit vector.

- ☐ True
☐ False

e.) Suppose A and B are 3×3 matrices. Then $\text{rank}(AB) = \text{rank}(A)$.

- ☐ True
☐ False

f.) If $\mathbf{u}, \mathbf{v}$ are linearly dependent vectors in $\mathbb{R}^2$, then $|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\|$.

- ☐ True
☐ False

h.) **(Multiple Choice-choose one)** If $T : P_2 \rightarrow \mathbb{R}^3$ is an isomorphism, which of the following is true?

- ☐ $\ker(T) = P_2$
- ☐ $\text{im}(T) = P_2$
- ☐ T is onto
- ☐ The inverse of T does not exist.
- ☐ None of the above.

i.) **(Multiple Choice-choose one)** Consider $\mathbb{R}^3$ with the standard inner product, and consider the subspace W of vectors in $\mathbb{R}^3$ which are orthogonal to both $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} -2 \\ 0 \\ 6 \end{bmatrix}$.

What is the dimension of W ?

- ☐ $\dim(W) = 0$
- ☐ $\dim(W) = 1$
- ☐ $\dim(W) = 2$
- ☐ $\dim(W) = 3$

j.) **(Multiple Choice-choose one)** Suppose A is a 3×3 matrix with eigenvalue 2 and eigenvalue 3. Which of the following is true?

- ☐ If $\mathbf{v}$ is in both the eigenspace of $\lambda = 2$ and the eigenspace of $\lambda = 3$, then $\mathbf{v} = \mathbf{0}$.
- ☐ If $\mathbf{w}$ is an eigenvector of A associated to $\lambda = 2$, then $-\mathbf{w}$ is an eigenvector of A associated to $\lambda = -2$.
- ☐ A is not diagonalizable.
- ☐ A is similar to I_3 .
- ☐ All of the above.

k.) **(Multiple Choice-choose one)** Suppose $A = \begin{bmatrix} -1 & x \\ -2 & y \end{bmatrix}$, and you know that A is similar to $\begin{bmatrix} -3 & -2 \\ 12 & 8 \end{bmatrix}$. Find x and y .

- ☐ $x = 6, y = 3$
- ☐ $x = -4, y = 22$
- ☐ $x = 12, y = 24$
- ☐ $x = 3, y = 6$
- ☐ None of the above.

4. (5 points total) Given $A = \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix}$, $C^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, and $D = \begin{bmatrix} 4 & 2 \\ 0 & 3 \end{bmatrix}$, find B if $(A + B)C = D$.
Show all work.

5. (22 points total) (This problem continues onto the next page)

Let $T : P_1 \rightarrow P_1$ be a linear transformation with $M_B(T) = \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix}$ with respect to the ordered basis $B = (x+3, x-1)$. (NOTE: B is NOT the standard basis for P_1).

a. (4 points) Compute $T(9-x)$. Show all work. (Note: your final answer should be in P_1).

b. (4 points) Find a basis for the kernel of T . (Note: your final answer should be in P_1).

c. (4 points) Is T diagonalizable? Justify your answer.

(Problem 5 continued) Again, let $T : P_1 \rightarrow P_1$ be a linear transformation with

$$M_B(T) = \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix} \text{ with respect to the ordered basis } B = (x+3, x-1).$$

d. (3 points) Is T an isomorphism? Justify your answer.

e. (3 points) Suppose $P_{B \leftarrow D} = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$ is the change matrix from unknown basis $D = (p_1, p_2)$ to B . Find p_1 and p_2 .

f. (4 points) For the same basis D as in part (e), find $M_D(T)$.
(Hint: You do NOT need to know D to solve this problem).

6. (7 points total) Consider the vector space P_1 , and define the function

$$\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$$

for $p(x), q(x)$ in P_1 .

You can assume that $\langle p(x), q(x) \rangle$ is an inner product for P_1 .

- a. (4 points) Using this inner product $\langle p(x), q(x) \rangle$, find the distance between $2x + 3$ and $x - 4$. Show all work. You do not need to simplify your final numerical answer.

- b. (3 points) Still using this inner product $\langle p(x), q(x) \rangle$, for which c is $cx + 2$, $2x - 2$ orthogonal? Show all work.

7. (7 points) Is $W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mid x_2 = 2x_1 + 3x_3, x_1x_3 \geq 0 \right\}$ a subspace of $\mathbb{R}^3$? If it is a subspace, prove it using the subspace test. If it is not a subspace, provide an explicit counterexample showing which of the subspace criteria is violated. Fully justify your answer.

8. (8 points total) Consider M_{22} , and let A, B be matrices in M_{22} . Let $\langle A, B \rangle = \text{tr}(AB^T)$. You can assume this is an inner product for M_{22} .

Consider $M = \begin{bmatrix} 6 & 1 \\ 2 & 4 \end{bmatrix}$ and $N = \begin{bmatrix} -1 & 2 \\ 2 & 0 \end{bmatrix}$. Consider $W = \text{span}\{M, N\}$.

- a. (4 points) Verify that M and N are orthogonal using this inner product. Show all work.

- b. (4 points) Assuming that $\{M, N\}$ is a basis for W , use M and N to build an orthonormal basis for W using this inner product. Show all work.

9. (7 points) Suppose a linear system of 3 equations and 3 unknowns (x, y, z) reduces to the following form via elementary row operations:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & (a^2 - 9) & (a + 3) \end{array} \right]$$

- (a) (2 points) For which value(s) of a will the system have **no solutions**? Briefly explain your answer.
- (b) (2 points) For which value(s) of a will the system have **infinitely many solutions**? Briefly explain your answer.
- (c) (3 points) If $a = 2$, finish reducing the augmented matrix to **reduced row echelon form**, then solve for x, y, z . Show all work.

10. (10 points) Suppose A is a 5×5 matrix with

$$RREF(A) = \begin{bmatrix} 1 & 0 & -2 & 0 & 2 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This is the reduced row echelon form of A , not A . Answer each question. Justification is not required.

a. (2 points) Is A invertible?

- ☐ Yes
☐ No
☐ Not enough information to determine.

b. (2 points) $\left\{ \begin{bmatrix} j \\ -1 \\ 0 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ k \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ is a basis for the nullspace of A . Find j and k . Write your answers in the provided boxes.

$j =$ $k =$

c. (2 points) Is 0 an eigenvalue of A ?

- ☐ Yes
☐ No
☐ Not enough information to determine.

d. (2 points) Is 1 an eigenvalue of A ?

- ☐ Yes
☐ No
☐ Not enough information to determine.

e. (2 points) If $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ is the linear transformation given by $T(\mathbf{x}) = A\mathbf{x}$, what is the dimension of the image of T ?

- ☐ 0
☐ 1
☐ 2
☐ 3
☐ 4
☐ 5
☐ Not enough information to determine.

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